THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM THREE MARCH MID – TERM EXAMINATIONS

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 3:00 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on every page of your answer booklet(s).

KSS BIO 1 F3 MID - TERM 2025

APR2025

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Form one students used a light microscope to study animal cells in the laboratory. They observed the organelle at the center of the animal cell which have tiny materials. The organelle is most likely to be the:

A Nucleus
B Cell membrane
C Chloroplast
D Vacuole
E Mitochondria

(ii) Which of the following plants lack xylem and phloem?

A Mushrooms and mosses
B Club mosses and horsetails
C Ferns and mosses
D Mosses and liverworts
E Maize and groundnuts

(iii) The basic principles of waste disposal include:

A Reducing, Burning and Incineration
B Reducing, Reusing and Recycling
C Recycling, Burning and Incineration
D Burning, Incineration and Tipping
E Reusing, Reducing and Tipping

(iv) What is the correct sequence of bones from your upper arm to the fingers?

A Scapula, humerus, ulna and radius, carpals and phalanges
B Humerus, radius and ulna, carpals, metacarpals and phalanges
C Clavicle, humerus, ulna and radius, carpals, metacarpals and phalanges
D Ulna, radius, biceps, triceps, scapula and phalanges
E Girdle, femur, radius and ulna, tarsals, metatarsals and phalanges

(v) We are insisted to abstain from unprotected sexual intercourse so as to avoid different diseases such as:

A Tuberculosis and heart failure
B Trichomoniasis and chlamydia
C Cholera and gastric ulcers
D Rickets and gonorrhea
E Marasmus and typhoid

(vi) Hypertonic and hypotonic solutions differ in that:

A Hypertonic solution has high water potential as compared to hypotonic solution
B Hypertonic solution loses water while hypotonic solution gains water by osmosis
C Hypertonic solution has low water potential as compared to hypotonic solution
D Both hypertonic and hypotonic solutions have equal water potential
E Both hypertonic and hypotonic solutions gain water from each other by osmosis

(vii) Conditions or factors that can change or be changed in an experiment are called:

A Hypothesis
B Variables
C Problems
D Conclusions
E Data

(viii) When students were doing private studies at night, light went off suddenly. Which of the following change occurred in the eyes of the students?

A Their eye lenses became thicker
B Their eyes opened wider
C Their eye irises enlarged
D Their eye pupils enlarged
E Their eyes produced tears

(ix) Examples of monocot plants include:

A Maize and sunflower
B Onion and garden peas
C Groundnuts and mango
D Sugarcane and millet
E Beans and tomatoes

(x) Incomplete treatment lowers the body immunity because:

A All pathogens are completely destroyed
B Pathogens are not completely destroyed
C Antibodies are produced by pathogens
D White blood cells accumulates in the body
E The medicine gets expired

2. Match the characteristics of living things in **List A** with their correct definitions in **List B** by writing its letter below the number of the corresponding item number in the booklet(s) provided.

LIST A	**LIST B**
(i) Growth	A The act of changing position
(ii) Movement	B Removal of metabolic wastes from the body
(iii) Nutrition	C Permanent increase is size and dry mass of an organism
(iv) Sensitivity	D Breaking down of food in living cells to produce energy
(v) Excretion	E The process by which an organism obtains food from the environment necessary for its survival
(vi) Respiration	F The act of responding to the stimuli
	G Giving rise to new ones of the same kind
	H The process of making food using sunlight

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Biology is not only about the study of plants and animals. It also deals with the study of many other organisms. Outline two examples of organisms studied in each of the following fields.
   (i) Entomology
   (ii) Microbiology

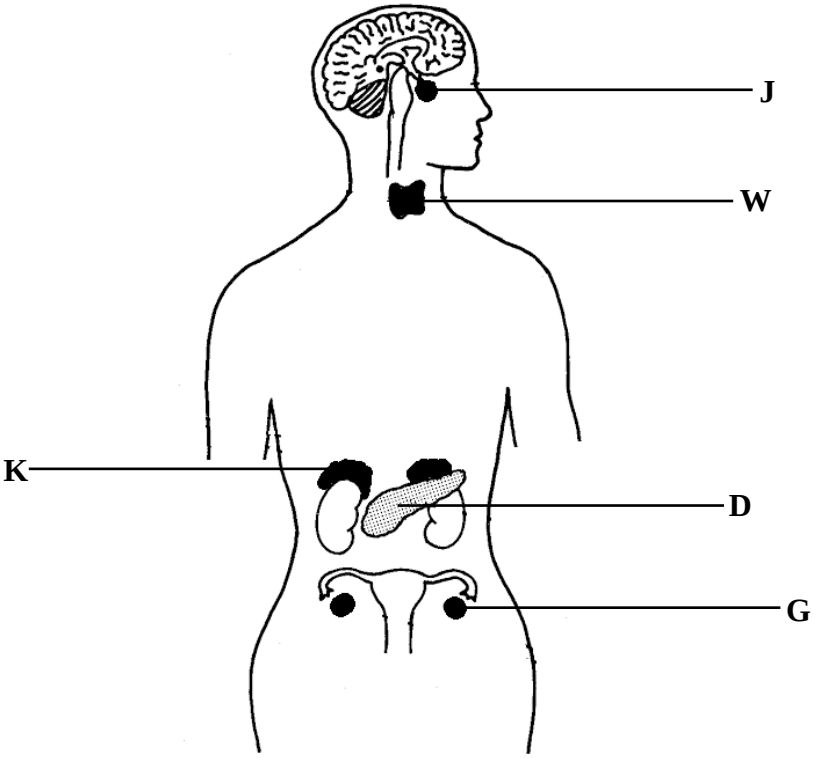
   (b) Bertha cooked rice and put it into a pot. She forgot to eat the rice and it remained in the pot for two weeks. Later, she opened the pot and found that the rice was covered by a mass of grayish structures which look like threads.
   (i) Give the common name of the structures that were observed on rice.
   (ii) State the kingdom and phylum of the structures named in (a) above.
   (iii) What is the mode of nutrition of these structures?
   (iv) Tabulate three key differences between plants and the structures grown on rice.

4. (a) As a form three biology student, describe in three points how you will protect your family members at home from being bitten by mosquitoes.
   (b) Maria and her friends were playing in the ground. Unfortunately, she fell down and got injured at her knee where she started bleeding. Her friends Agness and Monica put on gloves and helped her by washing her injuries, applying antibiotics and covering the injuries with a plaster.
      (i) Define the action done by Agness and Monica to their friend Maria.
      (iii) Why was it important for them to wear gloves before washing the injuries of Maria?
      (iv) Why did they apply antibiotics to the injuries?
      (v) What would happen to Maria if her friends would not help her? Provide three points.

5. (a) In which ways is transpiration necessary for plants? Provide three points.
   (b) Why do some people consider conifers as harmful plants in the environment?

6. (a) Distinguish between ecosystem and ecology.
   (b) With relevant examples from the environment, explain four forms of symbiotic relationships among living organisms.

7. Carefully study the figure below that shows different endocrine glands in the human body and then answer the questions that follow:



   (a) Give the name the glands labeled **G**, **K**, **D** and **W**.
   (b) Gland **J** is considered as 'the master gland'. Why?
   (c) With relevant examples, briefly explain how organ **D** functions as:
      (i) An endocrine gland
      (ii) An exocrine gland
   (d) When you see a snake, you would quickly gain unbelievable strength that will help you to either fight or run away. Describe how a hormone secreted by gland **K** will help you to get such strength. Provide four points.

8. (a) Draw a simple food guide pyramid and indicate appropriately how nutrients should be included in every meal to make it healthy and balanced.
   (b) Vanessa Girimeju weighs 5900grams on a beam balance and she measures 147centimetres in height.
      (i) What is her Body Mass Index (BMI)?
      (ii) According to the World Health Organization, what is the health status of Vanessa based on her BMI?
      (iii) State two health problems that Vanessa is likely to encounter due to her BMI.
      (iv) As a young biologist, provide a professional advice to Vanessa which corresponds to her health status. Give two points.

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. Explain the economic importance of anaerobic respiration in human's daily life.

10. Describe four types of movable joints found in the human body. Provide relevant example(s) for each type of joint.

11. Explain the health effects of drug abuse and hence propose the proper ways of handling drugs. Provide three points in each case.

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